

## 08 Geometric Transformations

### 8.2 Scaling

*Diagonal matrices and scaling transforms: uniform, non-uniform, squeeze, commutativity, negative and zero factors*

Linear Algebra Made Easy and Visual | Open-source code & drafts: <https://github.com/Visualize-ML>



## Core Skills You Will Master

1

Uniform scaling: diagonal matrix  $\mathbf{S}=s\mathbf{I}$ , determinant  $s^2$  (plane) or  $s^3$  (3D); the figure grows or shrinks as a whole

3

Horizontal/vertical-only scaling: scaling along a single axis while the other direction stays unchanged

5

Scaling matrix multiplication is commutative:  $\mathbf{S}_1 @ \mathbf{S}_2 = \mathbf{S}_2 @ \mathbf{S}_1$ , order does not affect the result

7

Zero scaling factor: the corresponding direction is projected away, the matrix is singular (rank drops)

2

Non-uniform scaling:  $\mathbf{S}=\text{diag}(s_1, s_2)$ ; understand  $\mathbf{A} @ \mathbf{S}$  (scaling  $\mathbf{A}$ 's columns) versus  $\mathbf{S} @ \mathbf{A}$  (scaling its rows)

4

Squeeze:  $s_1 s_2 = 1$  — one direction compresses, the other stretches, and area is preserved

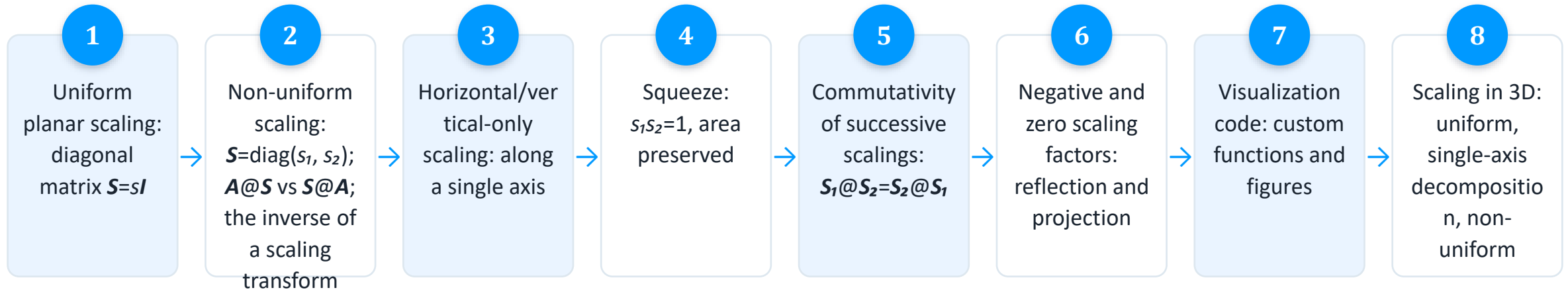
6

Negative scaling factor: the corresponding direction flips — a composite of “scaling + reflection”

8

Scaling in 3D: uniform, single-axis decomposition, and non-uniform — determinant tracks the volume ratio

# From Diagonal Matrices to the Full Picture of Scaling



## Seven Lenses for Every Transform in This Chapter

① Matrix-multiplication view: how the matrix acts on  $\mathbf{0}$ ,  $\mathbf{e}_1$ ,  $\mathbf{e}_2$ , and the all-ones vector  $\mathbf{1}$

② The determinant: is  $|\det|$  greater than 1? Positive, negative, or zero? Determinants of special matrices

③ How the determinant reflects the area or volume change caused by the transform

④ The inverse matrix (does it exist?) and the inverse transform; inverses of special matrices

⑤ Linear-combination view of the parallelogram/parallelepiped grid, and whether dimension drops

⑥ Basis view: does the transform matrix form a basis, and what kind of basis

⑦ Properties of the transform matrix: symmetric, diagonal, orthogonal, invertible, commutative

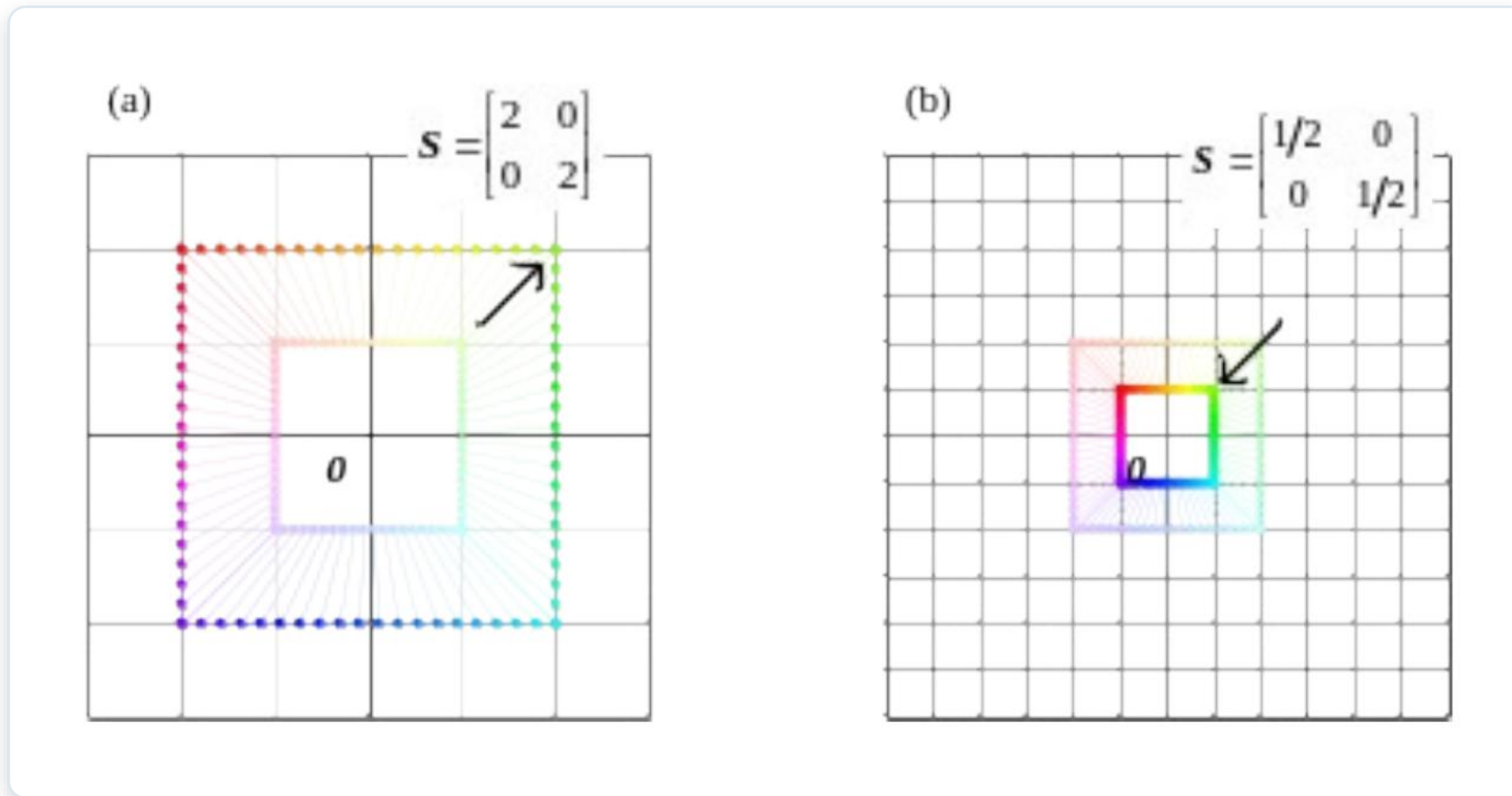
## Uniform Scaling: $S=sI$ , $s>1$ Enlarges, $0<s<1$ Shrinks

$$S = \text{diag}(s, s) = sI$$

- Uniform scaling enlarges or shrinks a figure by the same ratio in every direction;  $S$  has equal diagonal entries, like scalar multiplication
- $\det(S)=s^2$ , so the area changes by a factor of  $s^2$  after the transform; the columns of  $S$  form an orthogonal basis

## Figure 1 · Uniform Scaling, 2D Plane

- (a)  $s=2$  enlarges by 2 $\times$ ; (b)  $s=1/2$  shrinks by half — the two are mutual inverses; the origin never moves



## Vector $x$ Under Uniform Scaling Becomes $y=[sx_1, sx_2]$

$$[y_1, y_2]^T = S[x_1, x_2]^T = [sx_1, sx_2]^T$$

- If every diagonal entry of a diagonal matrix is nonzero, its inverse is also diagonal, with entries the reciprocals of the originals
- If two or more diagonal entries are 0, the matrix is singular — a preview of “zero scaling factor” later in this section

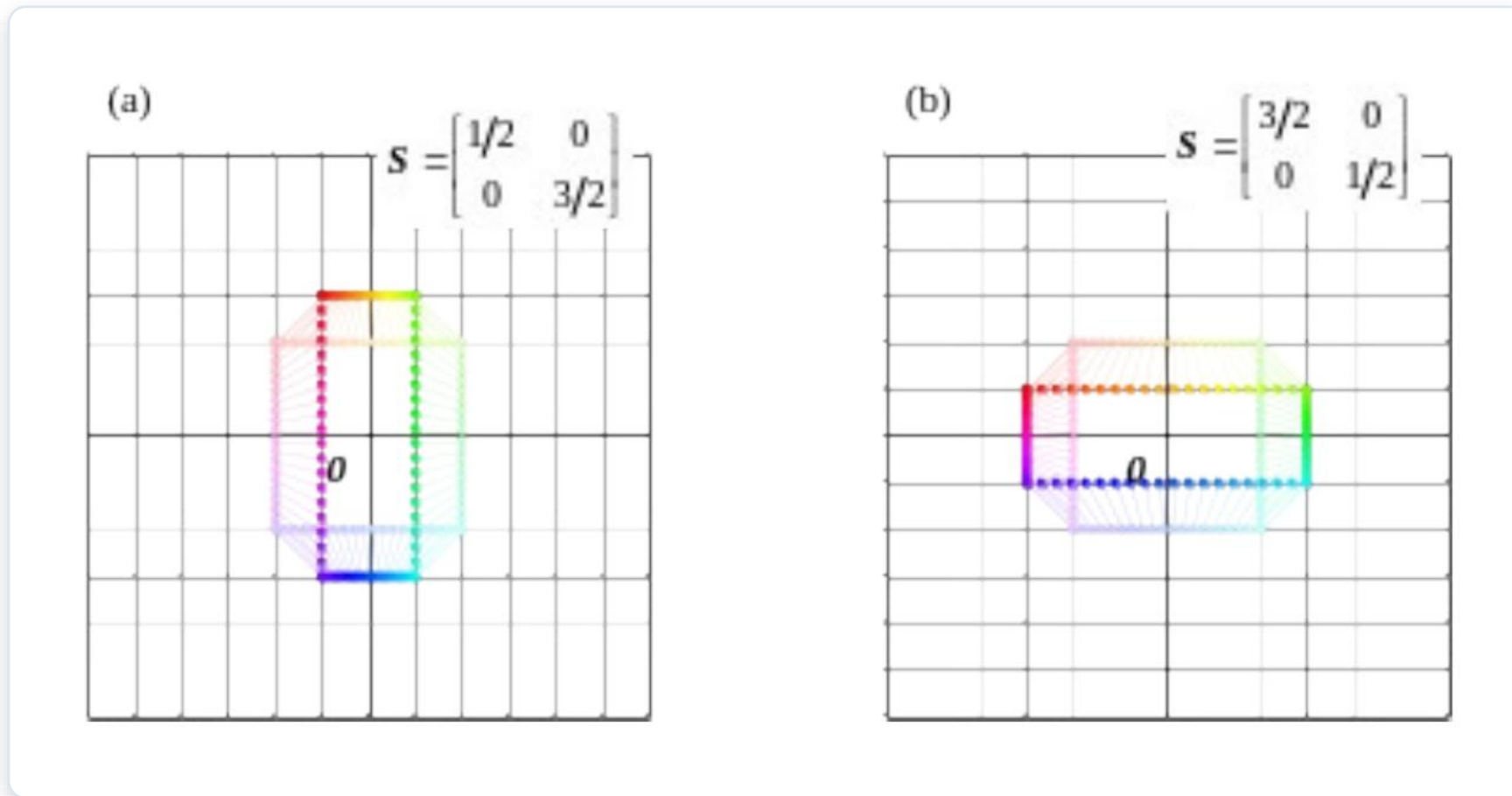
## $S = \text{diag}(s_1, s_2)$ : Different Ratios Horizontally and Vertically

$$S = \text{diag}(s_1, s_2)$$

- $s_1, s_2$  are both positive and  $s_1 \neq s_2$ ; a scaling factor of 1 means no change in that direction
- Recall: matrix multiplication  $\mathbf{A} @ \mathbf{S}$  scales the column vectors of  $\mathbf{A}$  by the diagonal entries of  $\mathbf{S}$  in turn

## Figure 2 · Non-uniform Scaling, 2D Plane

- (a)  $S = \text{diag}(1/2, 3/2)$ : horizontal shrink, vertical stretch; (b)  $S = \text{diag}(3/2, 1/2)$ : horizontal stretch, vertical shrink



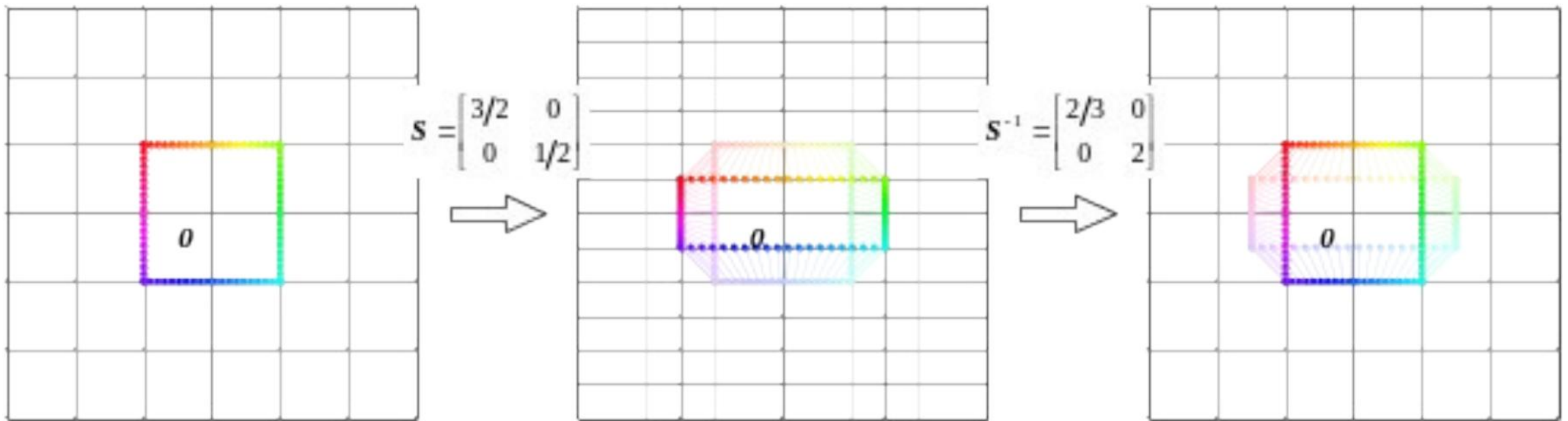
## $AS$ Scales $A$ 's Columns, $SA$ Scales Its Rows

$$AS = [s_1 \mathbf{a}_1 \quad s_2 \mathbf{a}_2] \quad SA = [s_1 \mathbf{a}^{(1)}; s_2 \mathbf{a}^{(2)}]$$

- For a  $2 \times 2$  matrix  $A$ ,  $A@S$  scales the column vectors  $\mathbf{a}_1, \mathbf{a}_2$  of  $A$  by the diagonal entries of  $S$  in turn
- $S@A$  scales the row vectors  $\mathbf{a}^{(1)}, \mathbf{a}^{(2)}$  of  $A$  by the diagonal entries of  $S$  in turn

## Figure 3 · Inverse of a Scaling Transform

- When  $\det(\mathbf{S}) \neq 0$ ,  $\mathbf{S}^{-1} = \text{diag}(1/s_1, 1/s_2)$  exists; if  $s_1$  or  $s_2$  equals 0,  $\mathbf{S}$  is not invertible



## $\det(\mathbf{S})=s_1s_2$ : Sign and Magnitude of the Area Ratio

$$\det(\mathbf{S}) = s_1s_2$$

- $|\det(\mathbf{S})|>1$  means the area of the figure grows;  $0<|\det(\mathbf{S})|<1$  means the area shrinks
- $\det(\mathbf{S})=0$  means dimension is lost and the matrix is singular — exactly the “zero scaling factor” case

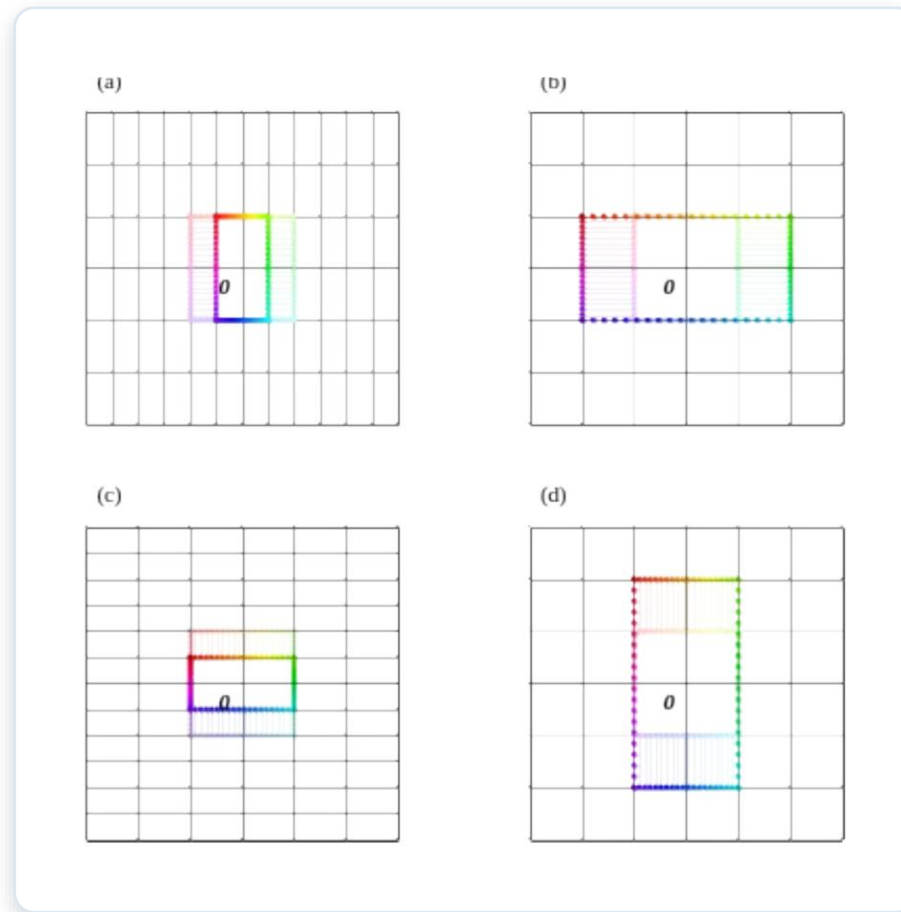
## Scaling Along a Single Axis, the Other Direction Unchanged

$$\mathbf{S} = \text{diag}(s, 1) \quad \mathbf{S} = \text{diag}(1, s)$$

- Horizontal-only scaling:  $\det(\mathbf{S})=s$ , so the area scales by a factor of  $s$  after the transform
- Vertical-only scaling follows the same form, with  $s$  moved to the second diagonal position

## Figure 4 · Horizontal and Vertical Scaling, 2D Plane

- (a)(b) two examples of horizontal-only scaling; (c)(d) two examples of vertical-only scaling



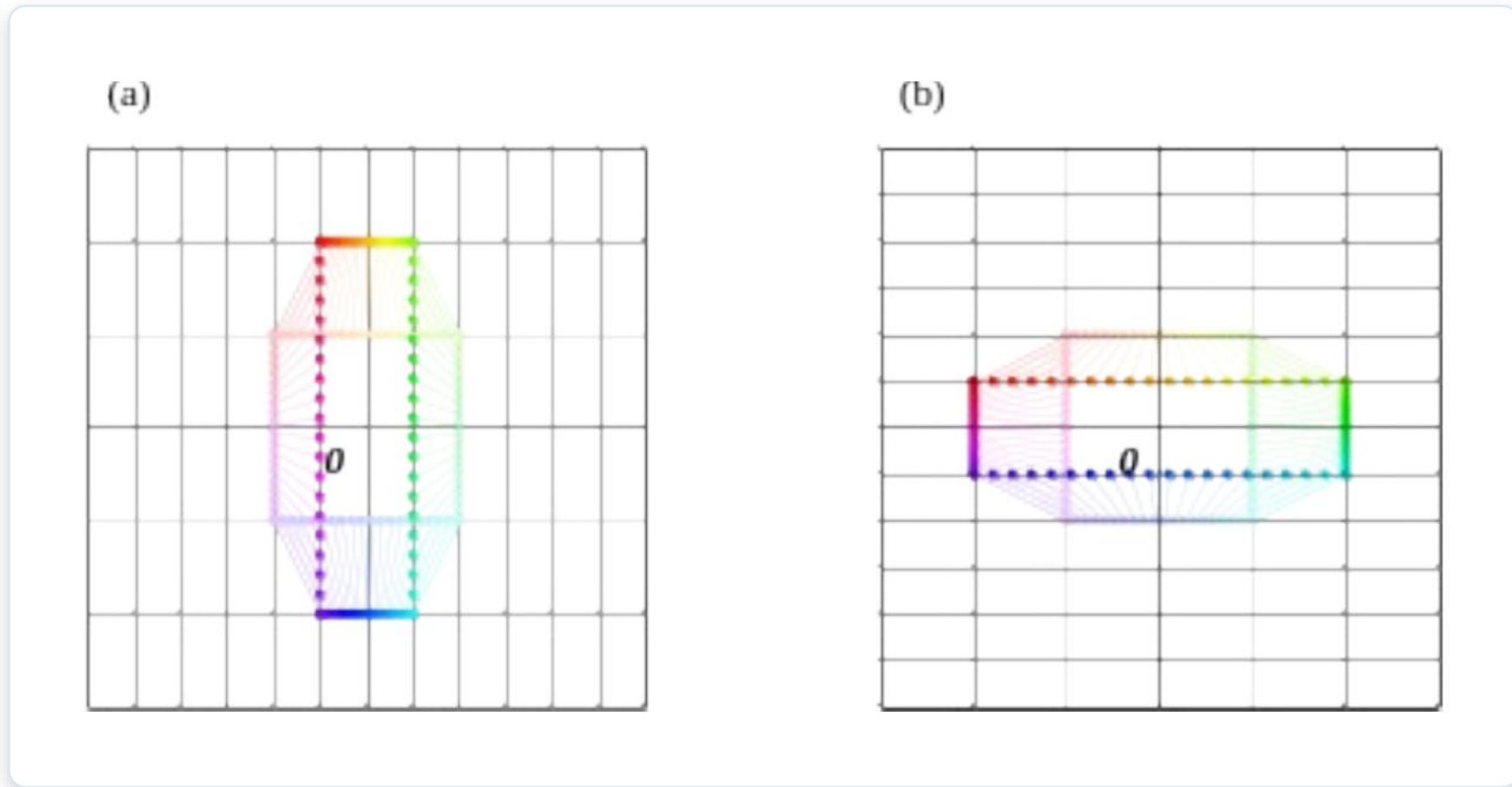
## Squeeze: $s_1 s_2 = 1$ , Area Is Preserved

$$s_1 s_2 = 1 \Rightarrow \det(\mathbf{S}) = 1$$

- A squeeze is a special non-uniform scaling: one direction has a factor below 1, the other above 1, and their product is 1
- For example  $0 < s_1 < 1$  and  $s_2 > 1$ : the figure compresses along  $x_1$  and stretches along  $x_2$ , with total area unchanged

## Figure 5 · Squeeze

- (a)  $s_1=0.5$  and  $s_2=2$ : horizontal shrink by half, vertical stretch by 2×, total area unchanged; (b) analyze it yourself



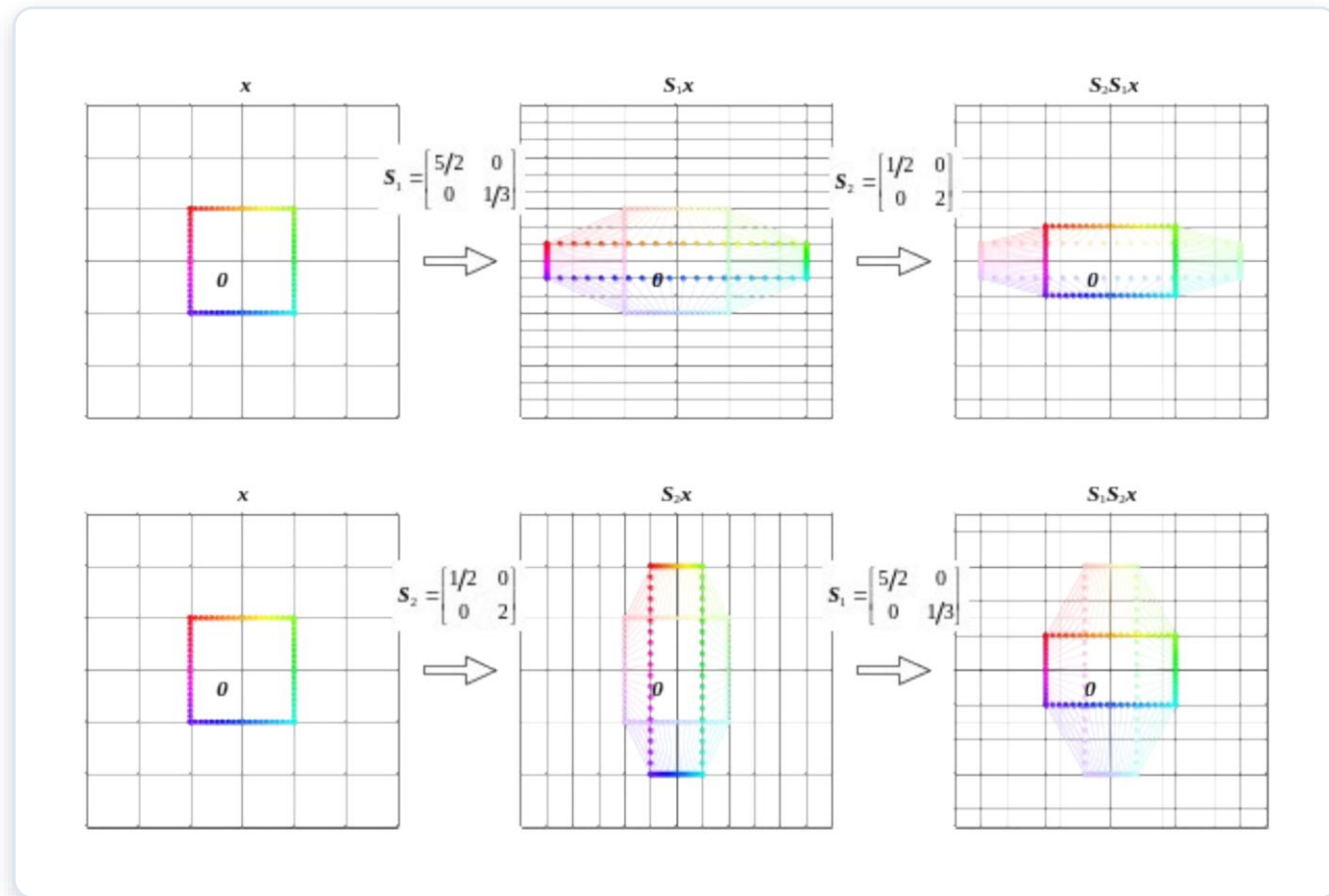
## A Special Case Where Matrix Multiplication Commutes

$$\mathbf{S} = \mathbf{S}_1\mathbf{S}_2 = \mathbf{S}_2\mathbf{S}_1$$

- Matrix multiplication generally does not commute, but a scaling matrix  $\mathbf{S}$  is diagonal and splits into two diagonal factors
- Geometrically, the horizontal and vertical scalings can be applied in either order; this holds for any two scaling matrices  $\mathbf{S}_1, \mathbf{S}_2$

## Figure 6 · Two 2×2 Scaling Matrices Commute

- Top row:  $S_1$  then  $S_2$ . Bottom row:  $S_2$  then  $S_1$ . Both orders produce exactly the same final result



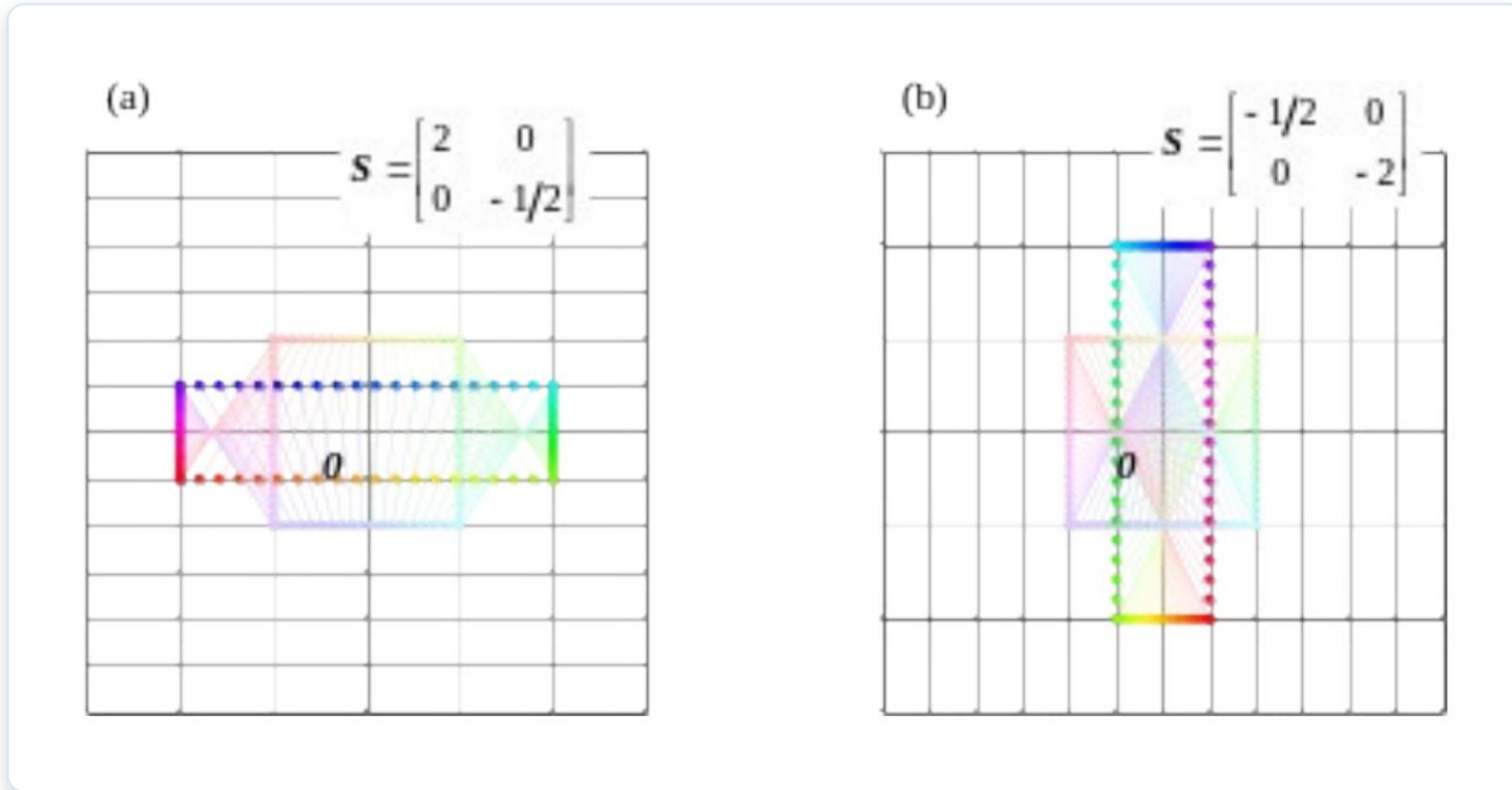
## Negative Scaling = “Scaling + Reflection”

$$\text{diag}(2, -1/2) = \text{diag}(1, -1) @ \text{diag}(2, 1/2)$$

- A negative scaling factor means the corresponding direction is flipped — a composite of “scaling + reflection”
- The equation above also shows that the order of the two operations does not affect the result

## Figure 7 · Scaling Factor With a Negative Sign

- (a)  $S = \text{diag}(2, -1/2)$ ; (b)  $S = \text{diag}(-1/2, -2)$  — analyze which directions flip in each case



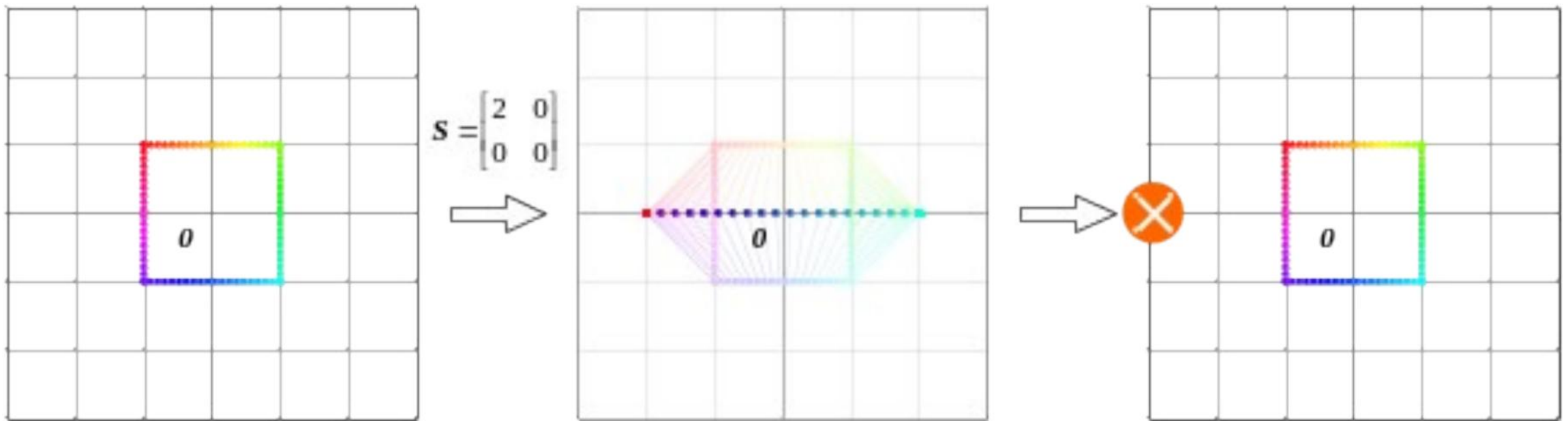
## Zero Scaling = “Scaling + Projection”, Information Is Lost

$$\text{diag}(0, 2) = \text{diag}(0, 1) @ \text{diag}(1, 2)$$

- A zero scaling factor means the corresponding direction is projected away — a composite of “scaling + projection”
- All points are projected onto the vertical axis, losing information;  $\det(\mathbf{S})=0$ , so  $\mathbf{S}$  is singular and its columns cannot form a basis

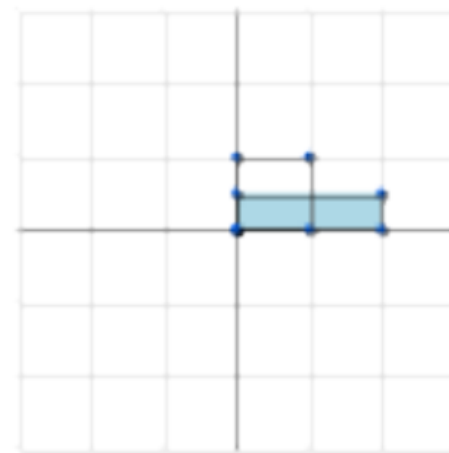
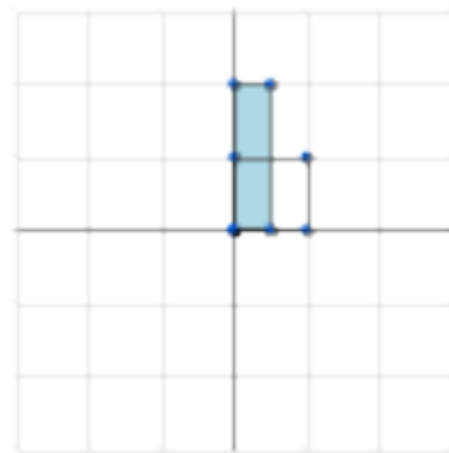
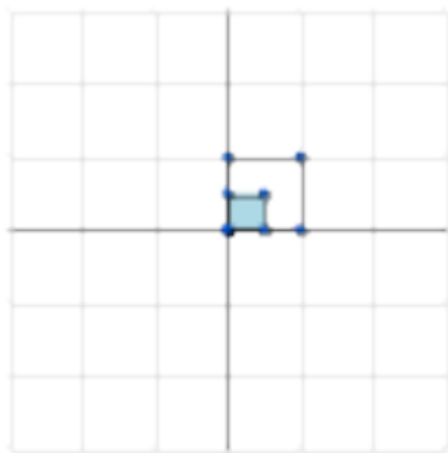
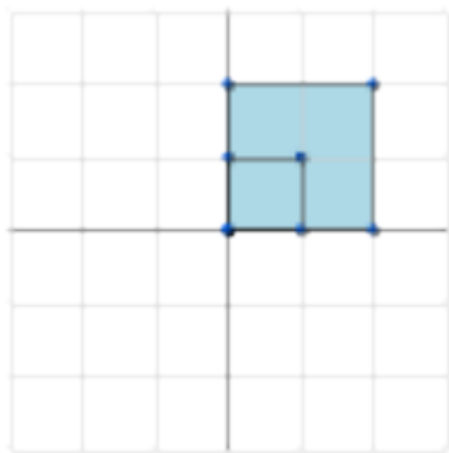
## Figure 8 · Irreversibly “Flattened”

- $S = \text{diag}(2, 0)$ : every point is flattened onto the horizontal axis; the red cross marks that no inverse exists



## Figure 9 · Visualizing Planar Scaling

- Four panels for four different scaling matrices: overall enlargement, uniform shrink, vertical-only scaling, horizontal-only scaling



## Custom Visualization: Planar Scaling

*The full pipeline for visualizing planar scaling*

- `numpy.array()` builds the linear transform matrix  $A$ ; the unit-square vertices form a  $5 \times 2$  array, each row a point on the plane
- $e_1, e_2$  are unit vectors along the horizontal and vertical axes;  $A$  performs the linear transform via matrix multiplication
- `ax.plot()` draws the original square (dashed) and transformed square (solid); `marker='o'` marks every vertex
- `ax.fill()` shades the transformed square light blue, making the scaling result easy to see

```
## 初始化
import numpy as np
import matplotlib.pyplot as plt

## 线性变换矩阵
a A = np.array([[2, 0],
                [0, 2]])

## 单位正方形顶点数据
b square_data = np.array([[0, 0],
                          [1, 0],
                          [1, 1],
                          [0, 1],
                          [0, 0]])

## 定义单位向量 e1 和 e2
c e1 = np.array([[1], [0]])
  e2 = np.array([[0], [1]])

## 线性变换
d transformed_square = square_data @ A.T
  transformed_e1 = A @ e1
  transformed_e2 = A @ e2

## 可视化
fig, ax = plt.subplots(figsize=(6, 6))

e ax.plot(square_data[:,0], square_data[:,1],
          c = 'k', lw = '0.5',
          marker = 'o', markerfacecolor = '#0066FF', ls = '--')
  ax.plot(transformed_square[:,0], transformed_square[:,1],
          c = 'k', lw = '0.5',
          marker = 'o', markerfacecolor = '#0066FF', ls = '--')
  ax.fill(transformed_square[:,0], transformed_square[:,1],
          color='lightblue')

ax.set_aspect('equal'); ax.grid(True, c = '0.8')
ax.axhline(y = 0, c = 'k'); ax.axvline(x = 0, c = 'k')
ax.set_xlim(-3, 3); ax.set_ylim(-3, 3)
```

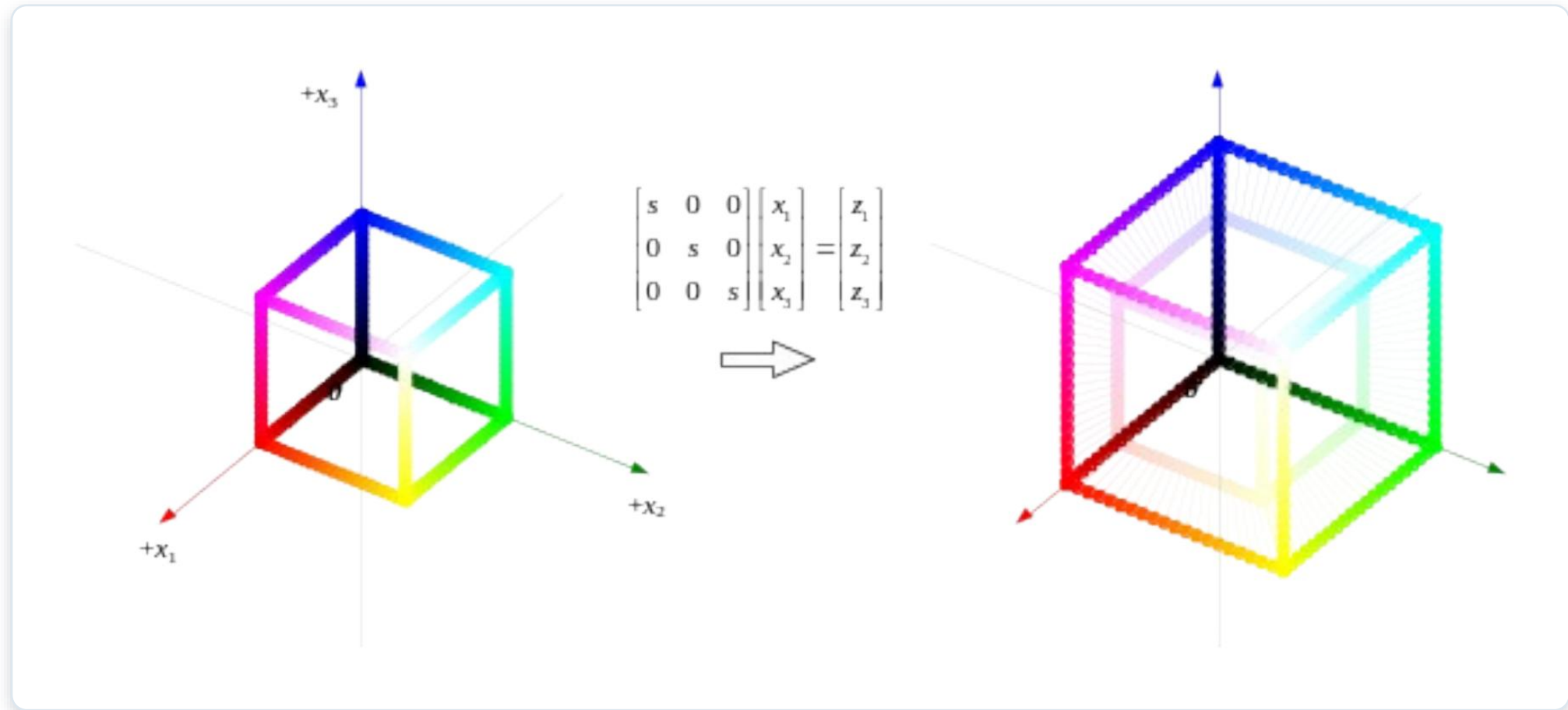
## Three Dimensions: $S=sI$ , $\det(S)=s^3$

$$S = \text{diag}(s, s, s) = sI$$

- $s$  is the scaling factor, nonzero; in 3D, uniform scaling applies the same factor  $s$  to all three dimensions
- The determinant of the transform matrix is  $s^3$ , so the volume changes by a factor of  $s^3$  after the transform

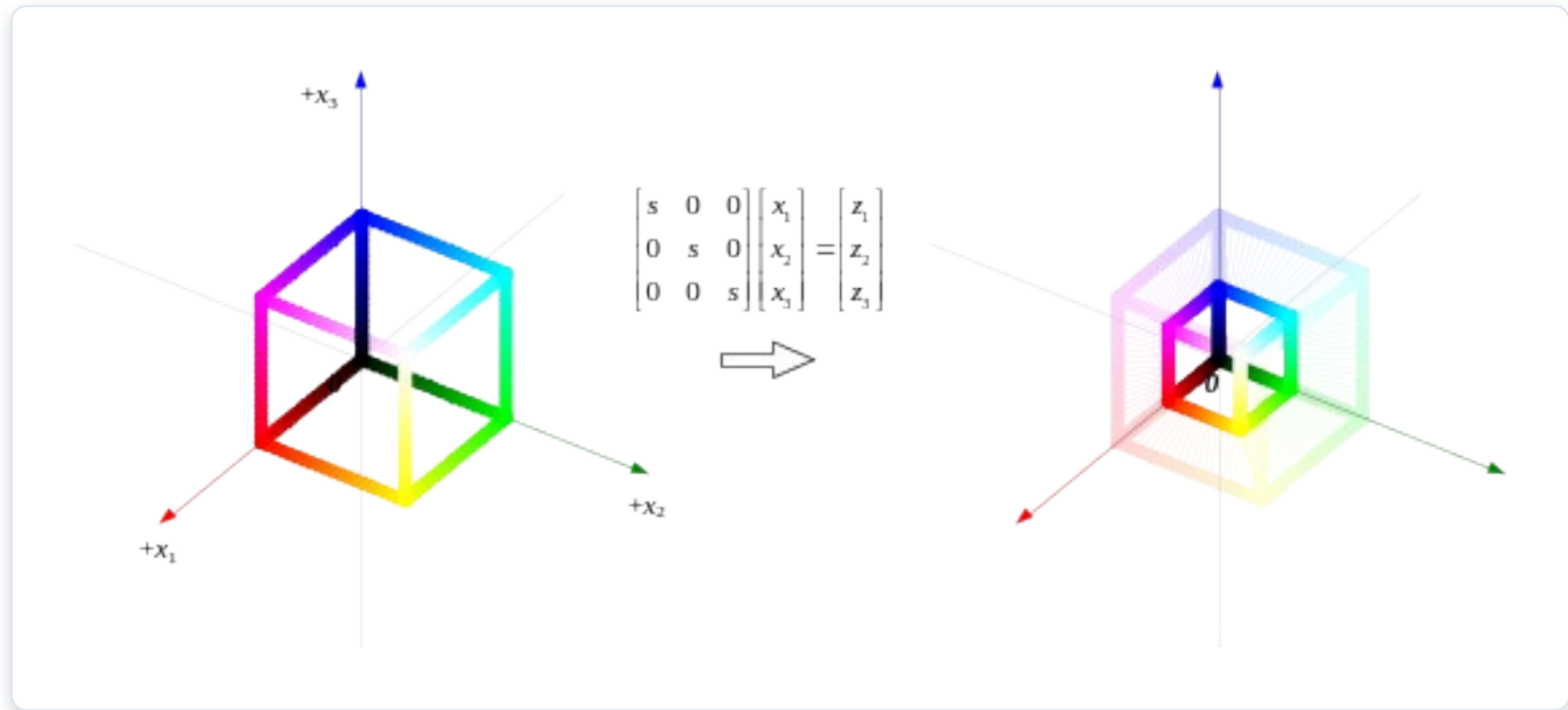
## Figure 10 · Uniform Enlargement ( $s > 1$ ), 3D Space

- The unit cube grows by the same ratio along all three axes at once; its shape is unchanged, only larger



## Figure 11 · Uniform Shrink ( $0 < s < 1$ ), 3D Space

- The unit cube shrinks by the same ratio along all three axes at once; its shape is unchanged, only smaller



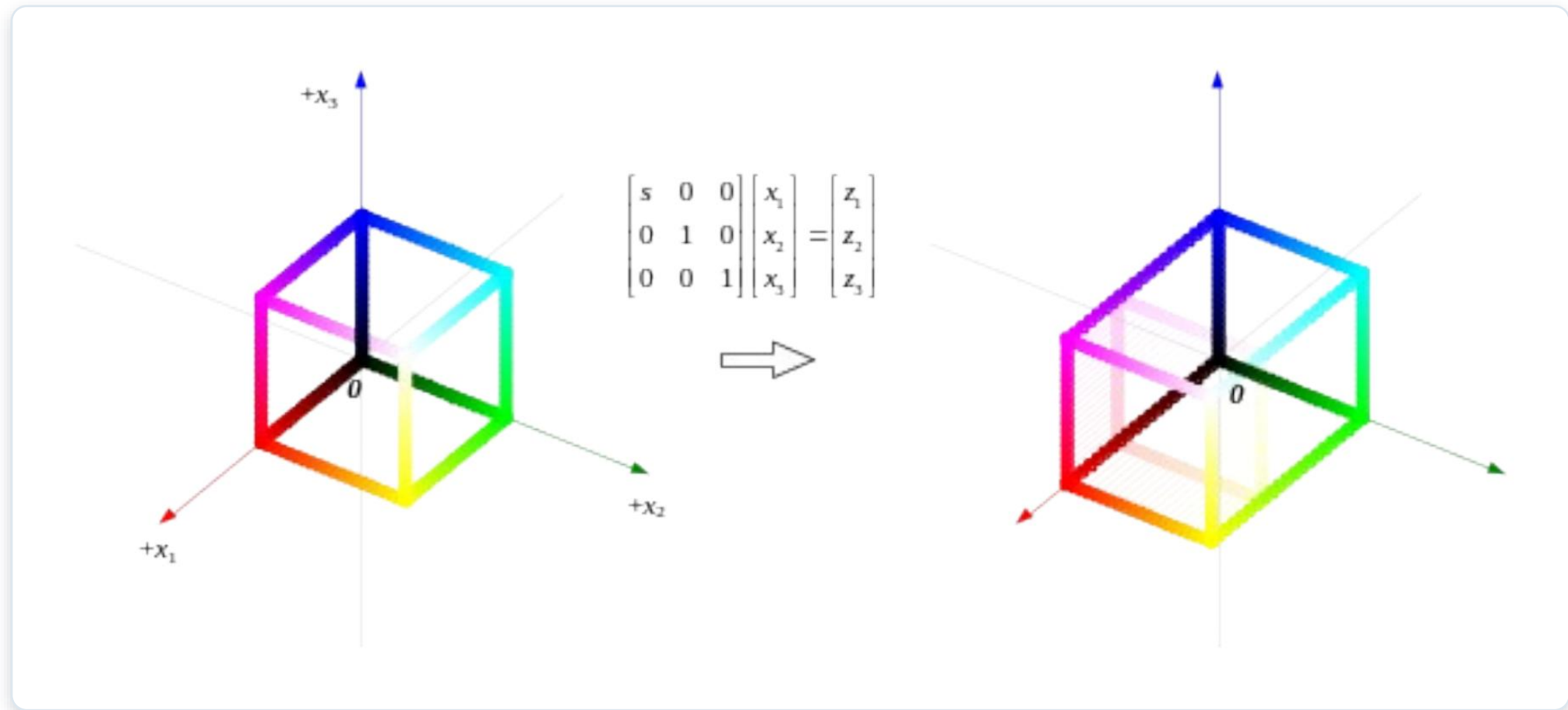
## $\mathbf{S} = \mathbf{S}_1 \mathbf{S}_2 \mathbf{S}_3$ : Splitting Into Scaling Along Each Axis

$$\mathbf{S}_1 = \text{diag}(s, 1, 1) \quad \mathbf{S}_2 = \text{diag}(1, s, 1) \quad \mathbf{S}_3 = \text{diag}(1, 1, s)$$

- Each of the three matrices has determinant  $s$ , so the volume scaling ratio caused by each is  $s$
- These three diagonal matrices give the same product  $\mathbf{S}$  regardless of the order they are multiplied in

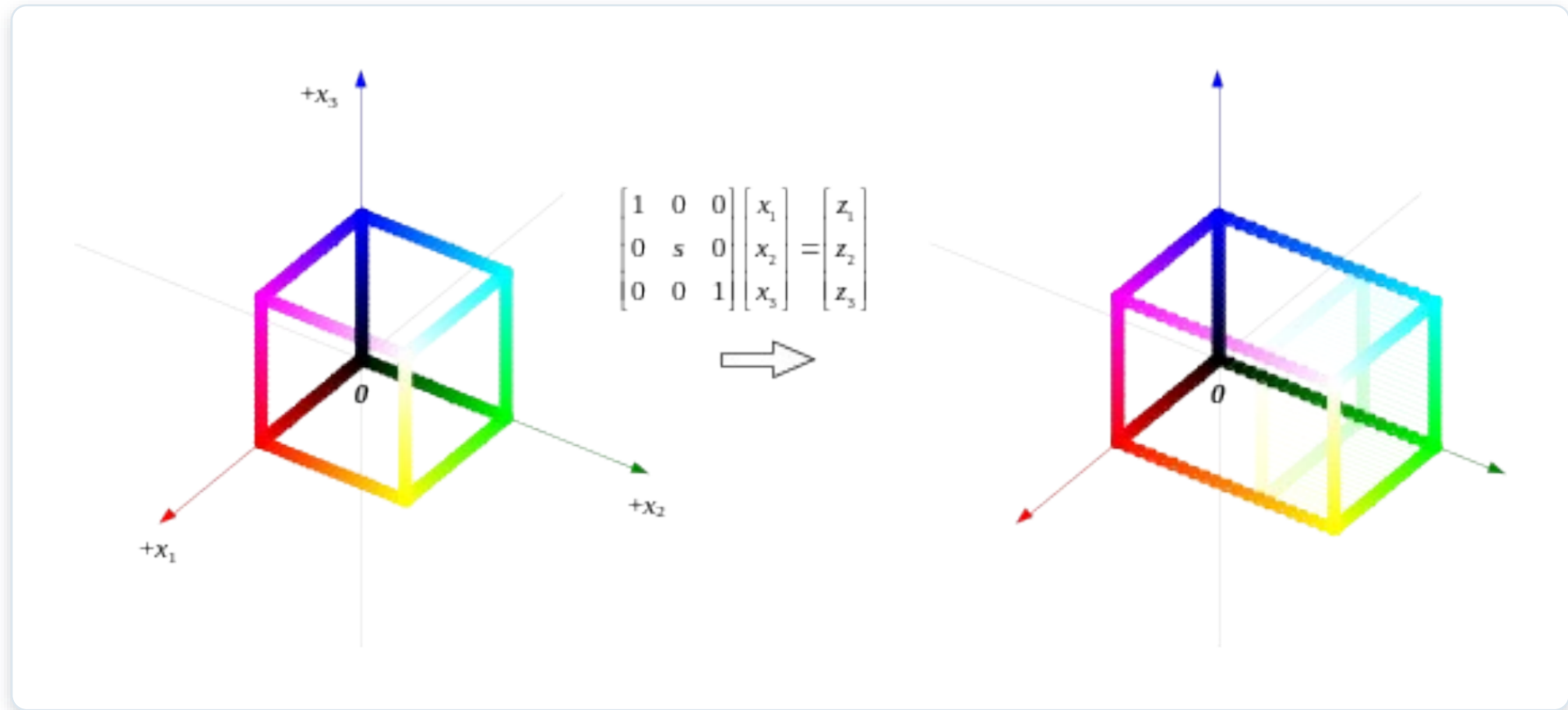
## Figure 12 · Scaling Along the $x_1$ Axis, 3D Space

- Only the edge length along  $x_1$  changes; the edge lengths along  $x_2$  and  $x_3$  stay the same



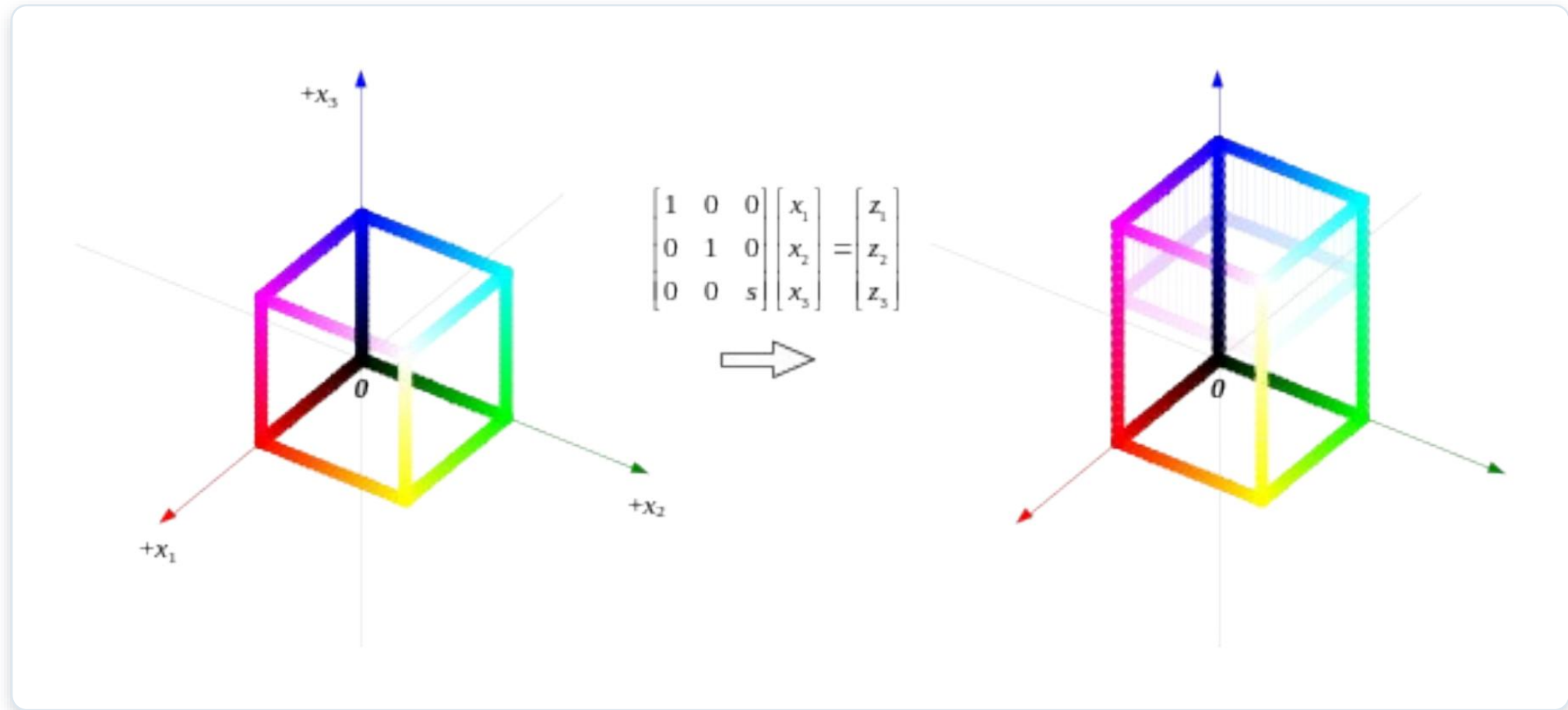
## Figure 13 · Scaling Along the $x_2$ Axis, 3D Space

- Only the edge length along  $x_2$  changes; the edge lengths along  $x_1$  and  $x_3$  stay the same



## Figure 14 · Scaling Along the $x_3$ Axis, 3D Space

- Only the edge length along  $x_3$  changes; the edge lengths along  $x_1$  and  $x_2$  stay the same



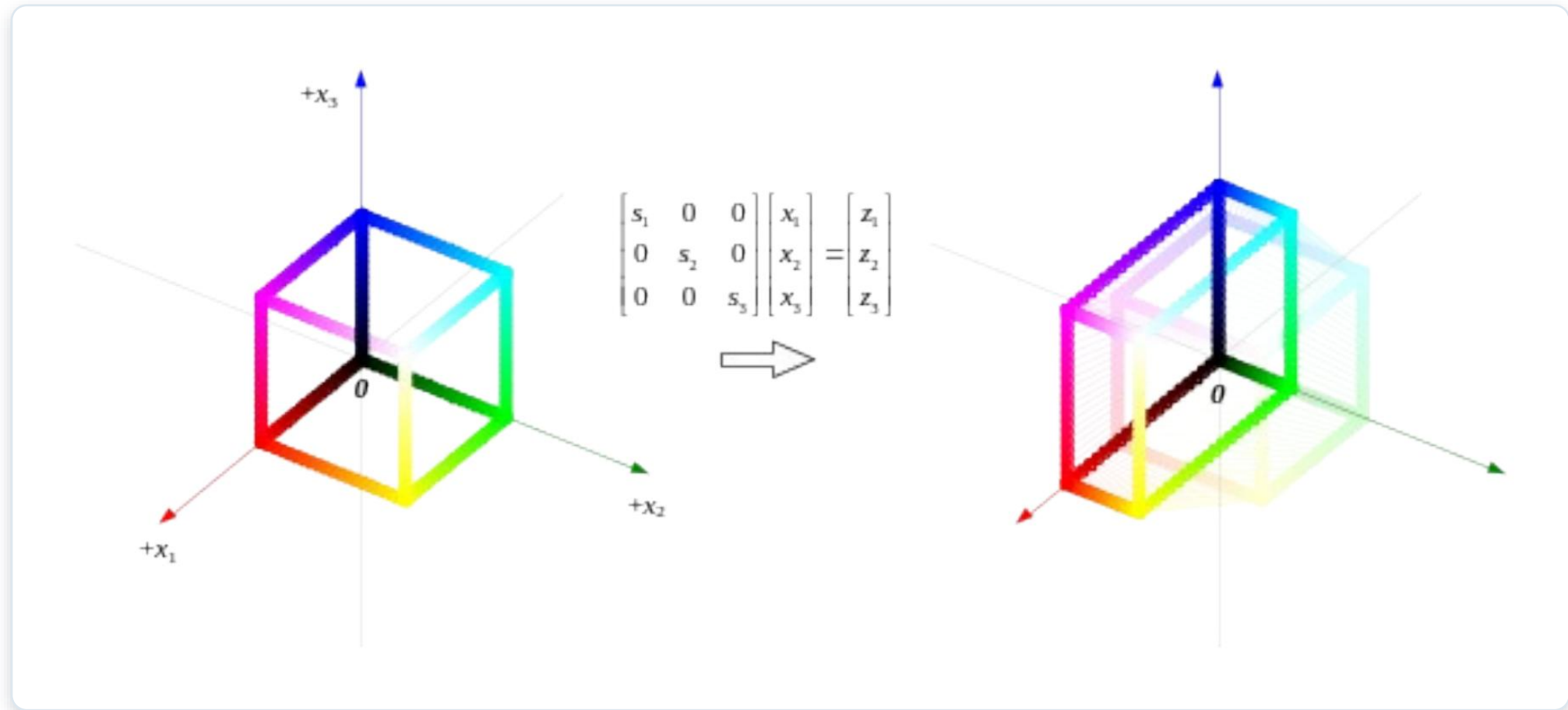
## $S = \text{diag}(s_1, s_2, s_3)$ : Each Axis Scales Independently

$$S = \text{diag}(s_1, s_2, s_3)$$

- $s_1$  controls scaling along  $x_1$ ,  $s_2$  along  $x_2$ , and  $s_3$  along  $x_3$
- A factor with  $|\text{value}| > 1$  enlarges that direction,  $|\text{value}| < 1$  shrinks it; negative means reflection too; 0 means “flattening”

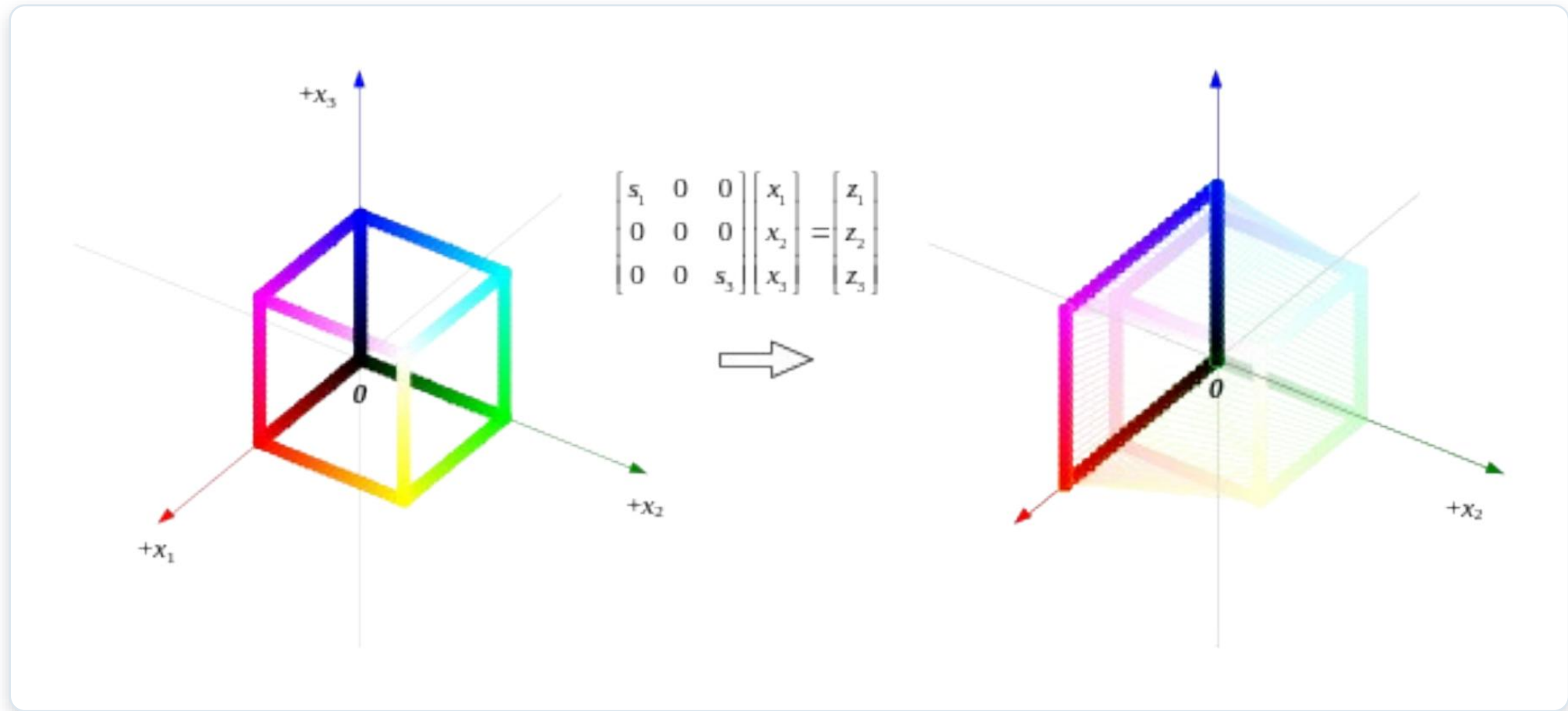
## Figure 15 · Non-uniform Scaling, 3D Space

- The scaling ratio differs along all three axes, turning the cube into a rectangular box



## Figure 16 · Scaling + Flattening, 3D Space

- When one scaling factor is 0, the figure is flattened into a plane along that direction — a loss of dimension, not invertible



## SUMMARY

# What Have We Seen From the Diagonal-Matrix View of Scaling?

Uniform vs non-uniform:  $\mathbf{S}=s\mathbf{I}$  scales everything together;  $\mathbf{S}=\text{diag}(s_1, s_2)$  scales each direction independently

Determinant:  $\det(\mathbf{S})$  is the area/volume ratio; its sign shows reflection, and 0 means singular (dimension lost)

Squeeze:  $s_1 s_2 = 1$ , area preserved; commutativity: diagonal matrices multiply in any order with the same result

Special cases: a negative factor = scaling + reflection; a zero factor = scaling + projection, matrix singular

## Key Terms in This Section

**Scaling Transform** 缩放变换 —— a linear transform that stretches or compresses a figure by a ratio

**Non-uniform Scaling** 不等比例缩放 —— different ratios per direction,  $S = \text{diag}(s_1, s_2)$

**Diagonal Matrix** 对角矩阵 —— a square matrix with zeros off the main diagonal

**Commutative Law** 交换律 —— diagonal matrices satisfy  $S_1 S_2 = S_2 S_1$

**Projection** 投影 —— a figure flattened onto a lower-dimensional subspace when a factor is 0

**Uniform Scaling** 等比例缩放 —— the same ratio in every direction,  $S = sI$

**Squeeze** 挤压 —— a special non-uniform scaling with  $s_1 s_2 = 1$ , area preserved

**Determinant** 行列式 —— reflects the area/volume ratio and orientation of a transform

**Reflection** 镜像 —— the flip that appears when a scaling factor is negative

**Dimensionality Reduction** 降维 —— determinant 0, transform not invertible, information lost

# Extended Reading

## Recap · 8.1 Geometric Transformations and Matrix Multiplication

Earlier in this chapter — five planar transforms through matrix multiplication

## Further Reading · 8.3 Rotation

Later in this chapter — rotation matrices, rotation angle, orthogonal matrices

## Further Reading · 8.5 Orthogonal Projection

Later in this chapter — orthogonal projection, Gram-Schmidt, QR decomposition

## Open-Source Repository

All code and figures for the book: <https://github.com/Visualize-ML>

## Q1 – Q9

**Q1** In the plane, a scaling transform enlarges the horizontal axis  $2\times$  and shrinks the vertical axis by  $1/2$ . What is  $S$ ?

**Q3** Given  $S=\text{diag}(-2, 2)$ , point  $(3, 4)$  is transformed by  $S$ . What are the new coordinates?

**Q5** If  $\det(S)=4$ , what does that tell you about the transform? What if  $\det(S)=-4$  instead?

**Q7** Given  $(1,0),(0,1),(-1,0),(0,-1),(0,0)$ , write Python code to scale them by  $s1=2$ ,  $s2=0.5$  and print the results.

**Q9** Preview: following the approach of the next section's LA\_08\_03\_01.ipynb, try visualizing scaling in 3D with code.

**Q2** In 3D, a scaling transform enlarges  $x_1$  by  $3\times$ , shrinks  $x_2$  by  $1/3$ , and leaves  $x_3$  unchanged. What is  $S$ ?

**Q4** Given  $S1=\text{diag}(1,2)$ ,  $S2=\text{diag}(3,1)$ , apply  $S1$  then  $S2$  to  $(1,2)$ . Swap the order — does the result change?

**Q6** Write a Python function `scale_2D(x1, x2, s1, s2)` that returns the scaled coordinates.

**Q8** Use Matplotlib to draw a unit square, scale it by  $s1=2$ ,  $s2=0.5$ , and show both the original and transformed figures.

## Q10 – Q17

**Q10** Use Python to verify: multiply squeeze matrices  $\text{diag}(0.5, 2)$  and  $\text{diag}(2, 0.5)$ , and check whether order matters.

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**Q12** Use numpy to check whether  $\text{diag}(3, 1/3)$  times  $\text{diag}(1/3, 3)$  is the identity, and whether order matters.

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**Q14** Write a Python function `has_reflection(s1,s2)` that checks whether either scaling factor is negative.

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**Q16** Write a Python function `is_singular_scaling(s1,s2)` that checks whether  $\text{diag}(s1, s2)$  is not invertible.

**Q11** Write a Python function `is_squeeze(s1,s2)` that checks whether two scaling factors form a squeeze.

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**Q13** Construct two scaling matrices  $S1$ ,  $S2$  and verify with Python that  $S1@S2=S2@S1$  always holds for diagonal matrices.

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**Q15** For  $S=\text{diag}(-3, 2)$ , compute the sign and the absolute value of the determinant, and explain each geometrically.

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**Q17** For  $\text{diag}(0, 5)$ , use Python to verify which axis all points get projected onto, and visualize the result.

## Q18 – Q25

**Q18** Write a Python function `scale_3D(x1,x2,x3,s1,s2,s3)` that returns the non-uniformly scaled 3D coordinates.

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**Q20** Given a  $3 \times 3$  non-uniform scaling matrix, write Python code to compute its determinant and decide if volume grows or shrinks.

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**Q22** Use Python to compare  $S^{-1}$  with taking reciprocals of the diagonal entries, and find when the inverse fails to exist.

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**Q24** Use Matplotlib to visualize a unit square under positive, negative, and zero scaling factors, and compare the results.

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**Q19** Use numpy to verify: three 3D scaling matrices  $S_1, S_2, S_3$  give the same product regardless of multiplication order.

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**Q21** Construct a  $3 \times 3$  scaling matrix with one factor equal to 0, and verify the transformed unit cube has zero volume.

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**Q23** Write a Python function `compare_AS_SA(A, s1, s2)` that compares right- and left-multiplying  $A$  by a scaling matrix.

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**Q25** Preview thought: does the commutative law hold the same way for diagonal matrices and the orthogonal matrices of rotation?

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# 10 Prompts to Explore With AI

1 Explain why diagonal matrices  $\mathbf{S}_1, \mathbf{S}_2$  commute under multiplication while matrix multiplication generally does not.

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2 Using Python, compare  $\mathbf{A}@\mathbf{S}$  with  $\mathbf{S}@\mathbf{A}$  (where  $\mathbf{S}$  is a diagonal scaling matrix) for a given matrix  $\mathbf{A}$ .

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3 Explain why a squeeze transform preserves area, and prove it using the determinant.

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4 Explain why a negative scaling factor is equivalent to a composite of “scaling + reflection”.

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5 Explain why a zero scaling factor makes a matrix singular, and how this relates to the matrix's rank.

# 10 Prompts to Explore With AI

6 Using Python, verify that three matrices scaling along the three 3D axes give the same product in any order.

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7 Explain the relationship between the absolute value of a determinant and the area/volume ratio of a transform.

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8 Design an experiment comparing uniform and non-uniform scaling applied to a unit circle.

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9 Explain why the inverse of a scaling matrix takes reciprocals of the diagonal entries, and when it fails to exist.

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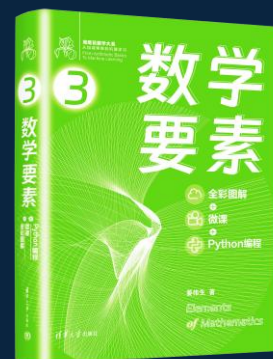
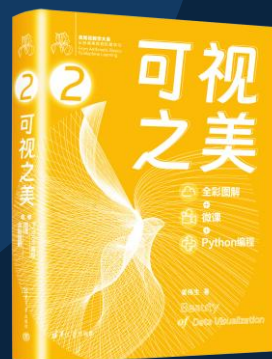
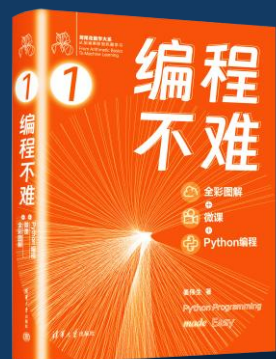
10 Compare scaling with the rotation transform in the next section: both are linear, but which properties does each preserve?

# 数学

数学基础

线性代数

概率统计



Python编程

数据可视化

工具

数据分析  
图和网络

回归、降维  
分类、聚类

实践

